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A Game-Theoretic Framework for Hydrogen Integration to Accelerate Energy Transitions in Great Britain

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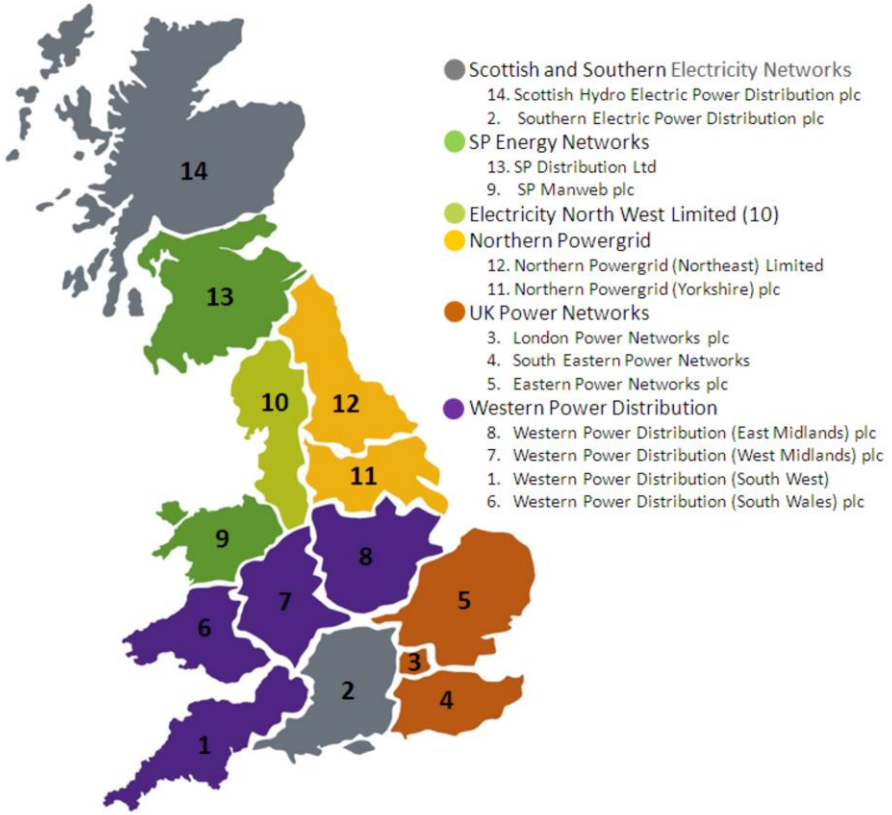
A Game-Theoretic Framework for Hydrogen Integration to Accelerate Energy Transitions in Great Britain

Objective:

To assess the optimal strategies for decarbonizing the GB's energy system, with focus on hydrogen integration.

Research Questions:

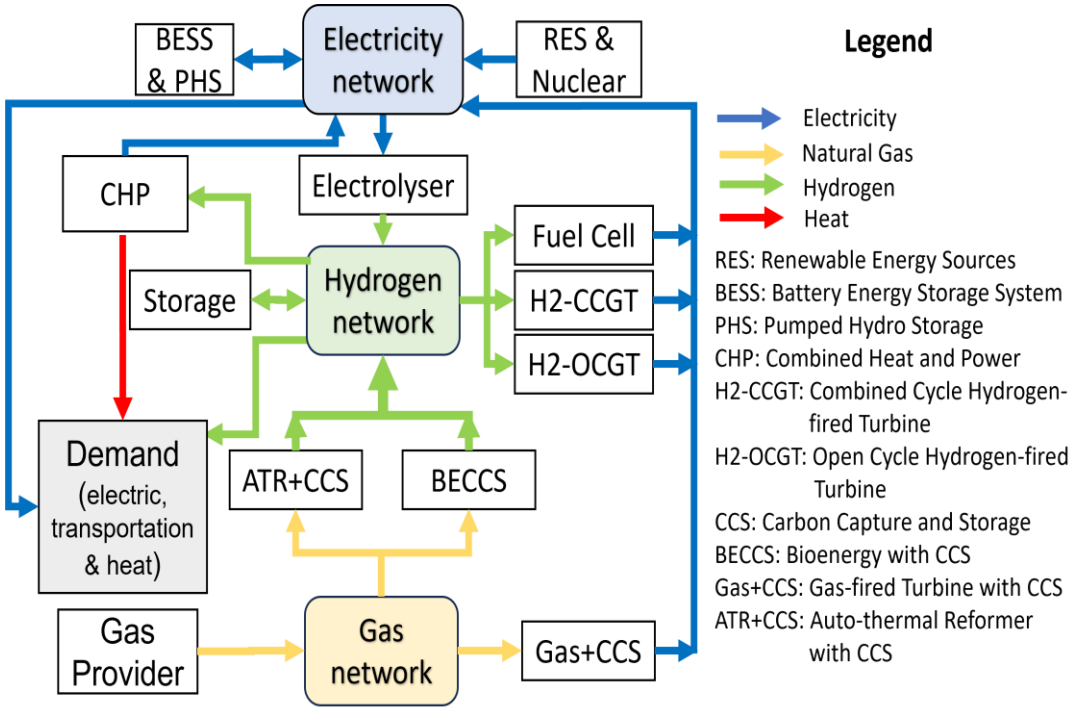
1. What are the optimal strategies for incorporating hydrogen into existing energy systems?
2. How can hydrogen contribute to decarbonisation across electricity, heating, and transportation?
3. What are the economic and technical challenges of hydrogen integration?
4. How can optimisation models address these challenges and identify pathways for hydrogen integration?



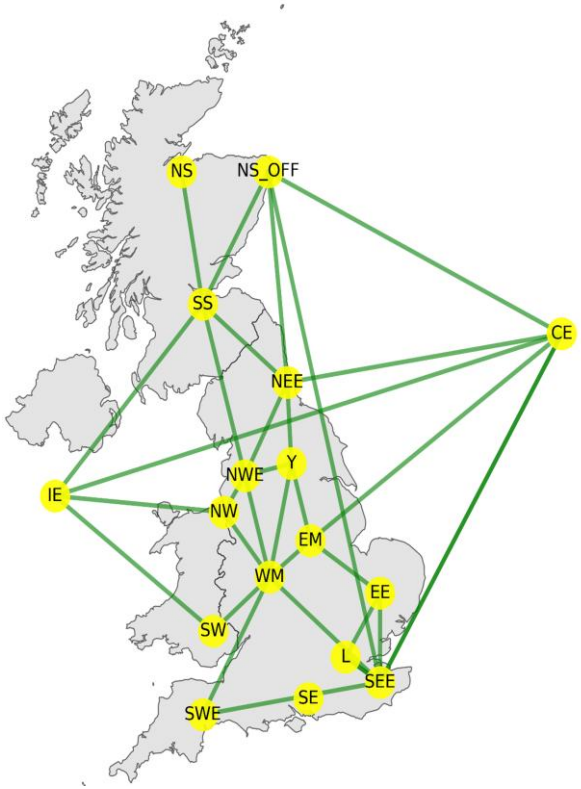
Zones of the DNOs in the GB

System Description:

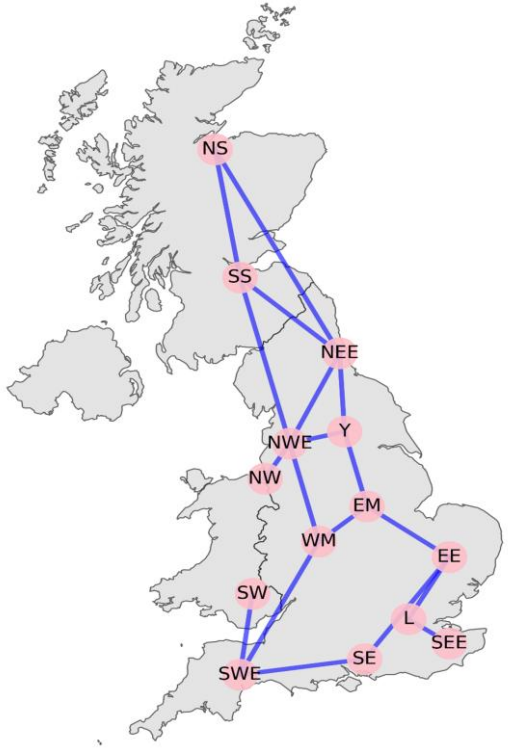
The GB's energy systems networks: Electricity and gas/hydrogen networks have been built using pandapower and pandapipes, and Pyomo in Python.



Integrated multi-vector energy system framework



Electricity Network



Gas/Hydrogen Network

Methodology: Modelling Long-term Planning & Short-term Operation

Forms the foundational optimization layer for the multilevel optimization framework, which incorporates short-term operation (OPGF) and long-term planning (game-theoretic model) to achieve GB's net-zero by 2050.

We solve for a Nash-Cournot Equilibrium, where no single player can improve their outcome by unilaterally changing their investment strategy.

Each player's decision is driven by maximizing their Net Present Value (NPV).

$$\text{NPV} = AF\left(\sum_{t=0}^{T}(R_t - C_t) - FC\right) - NI$$

where:

$$AF = \frac{(1+r)^n - 1}{r(1+r)^n}$$

r = Discount rate, n = years of the planning horizon

R_t = Revenue at time t

C_t = Opex cost at time t (£)

$$FC = \sum_{z \in Z}\left(\phi_z \cdot \varepsilon_z \cdot (K_z + I_z)\right)$$

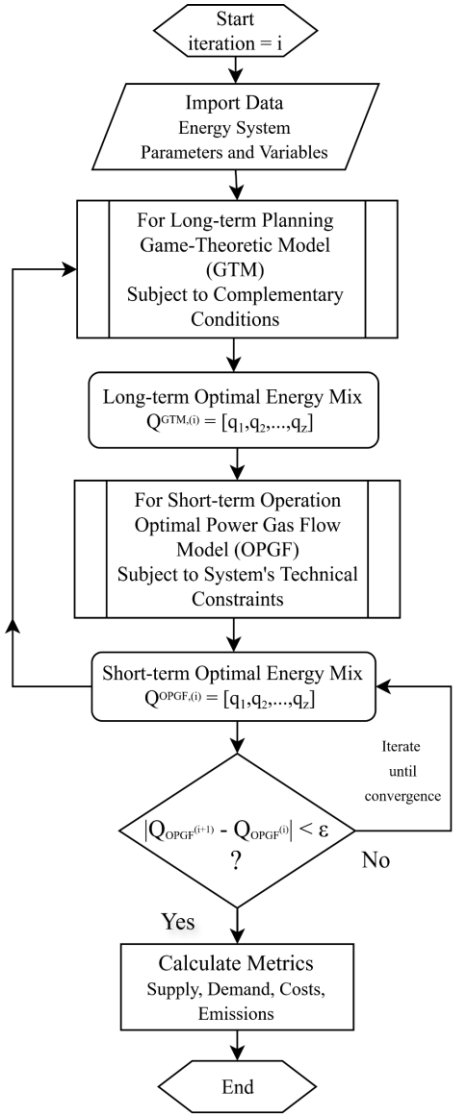
ϕ_z = Fixed cost fraction of technology z (%)

ε_z = Capex cost of technology z (£/MW)

K_z = Initial capacity of technology z (MW)

I_z = Invested capacity of technology z (MW)

NI = Net investment outlays (£)



Framework Flowchart

Simulation

Installed capacity settings for four investment scenarios in 2050, with 2025 as baseline capacities.

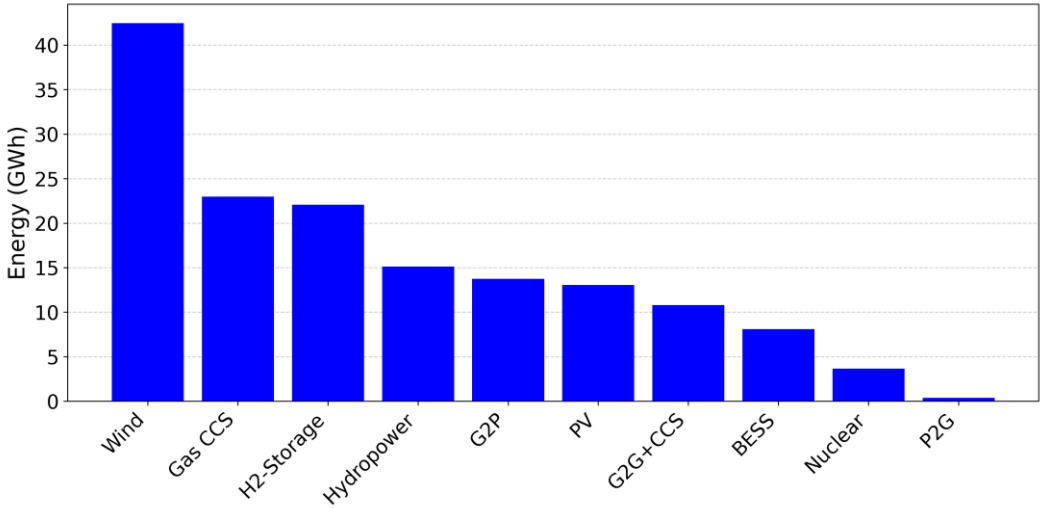
The electric peak demand is 102.17GWh, while hydrogen peak demand=14.27 GWh.

Technology	**2025** (GW)	**2050 Scenarios** (GW)			
		Uniform +25GW/Tech.	High RES–High H_2	High RES–Low H_2	High H_2–Low BESS
Nuclear	6.8	31.8	25	25	25
Wind (offshore, onshore)	31	81	100	100	150
Solar (PV)	17	42	70	70	110
BESS	3	28	60	60	10
Others	>25	>25	>25	>25	>25
G2P (H2-CCGT)	0	25	10	0.5	25
G2P (H2-OCGT)	0	25	10	0.5	25
G2P (Fuel Cell)	0	25	10	0.5	25
P2G (Electrolyser)	0.5	25.5	20	0.5	25
G2G (ATR+CCS, BECCS)	0	25	15	0.5	25

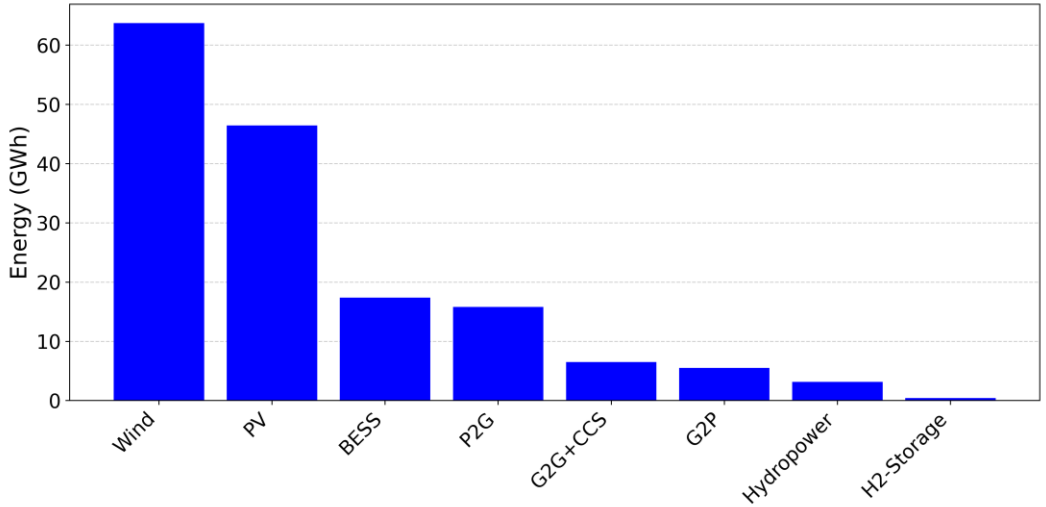
Results

2025-2050 Planning

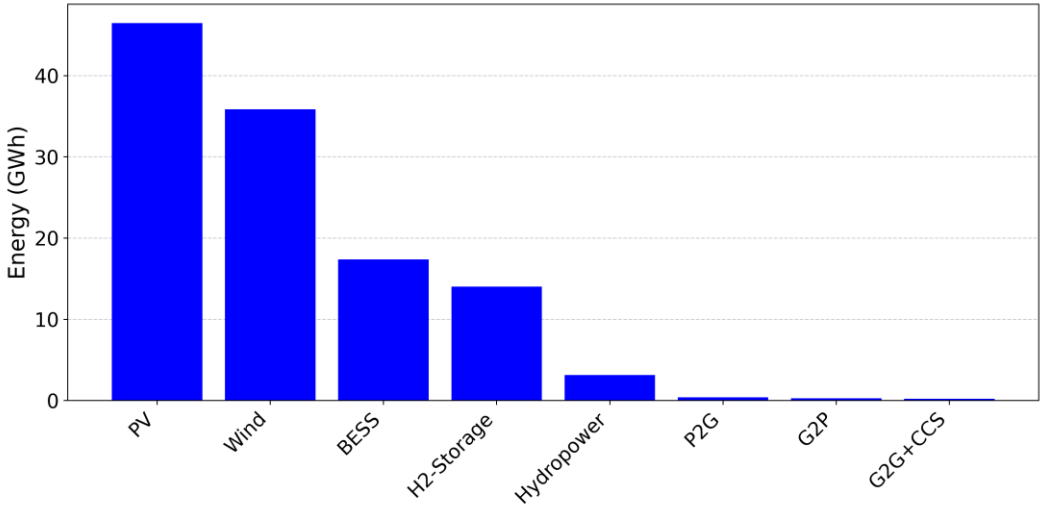
The electric peak demand is 102.17GWh, while hydrogen peak demand=14.27 GWh.



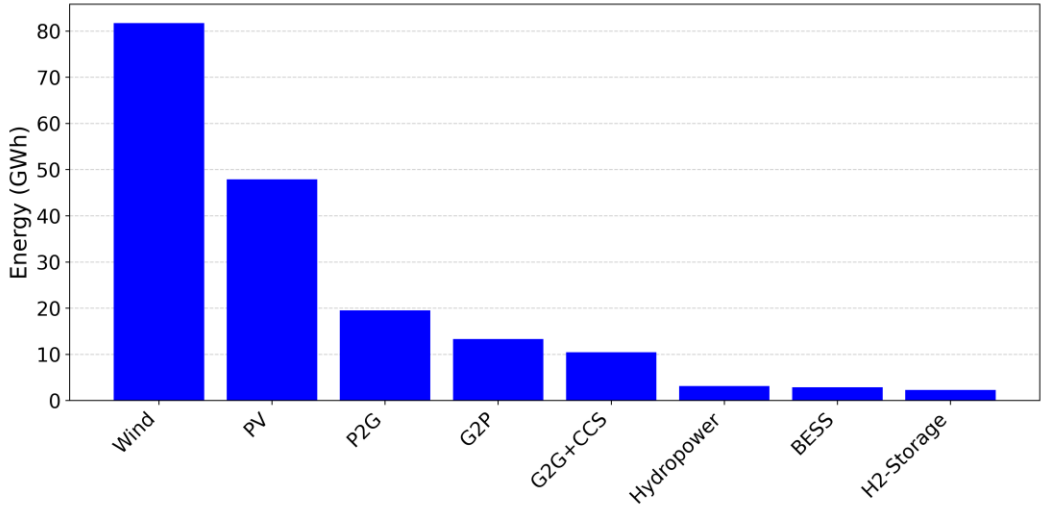
Uniform +25GW/Tech.



High RES–High H_2



High RES–Low H_2



High H_2–Low BESS

Results

Simulation results for 2050 peak demand under different capacity allocation scenarios.

The electric peak demand is 102.17GWh, while hydrogen peak demand=14.27 GWh.

Metric	Uniform +25GW/Tech.	High RES–High H_2	High RES–Low H_2	High H_2–Low BESS
Total Demand (GWh)	152.51	158.1	117.75	182.75
Electric Demand (GWh)	119.2	136.22	103.1	149.44
Hydrogen Demand (GWh)	33.31	21.88	14.65	33.31
Total Supply (GWh)	152.51	158.1	117.75	182.75
Operational Cost (£m)	6.64	3.61	2.05	4.76
CO_2 Emissions (tCO_2eq)	734	0	0	0
G2P / Peak Demand (%)	13.46%	5.38%	0.27%	13.02%
RES Utilisation (%)	45.15%	64.82%	48.41%	43.22%

Conclusion

- The developed bi-level game-theoretic framework offers actionable insights for decarbonisation planning;
- Hydrogen as key flexibility vector:
  - ✓ Hydrogen can supply up to 13.02% of electric peak demand;
  - ✓ Achieved 64.82% renewable utilisation, reducing curtailment and compensating for limited batteries;
- Operational costs range: £2.05m – £6.64m, showing infrastructure scale and cost–flexibility trade-offs;
- Policy support (e.g., electrolyser subsidies) is essential for hydrogen feasibility;
- Future work: cooperative game theory for coalition-based planning.

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Thank You for Your Attention

Any Questions?

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